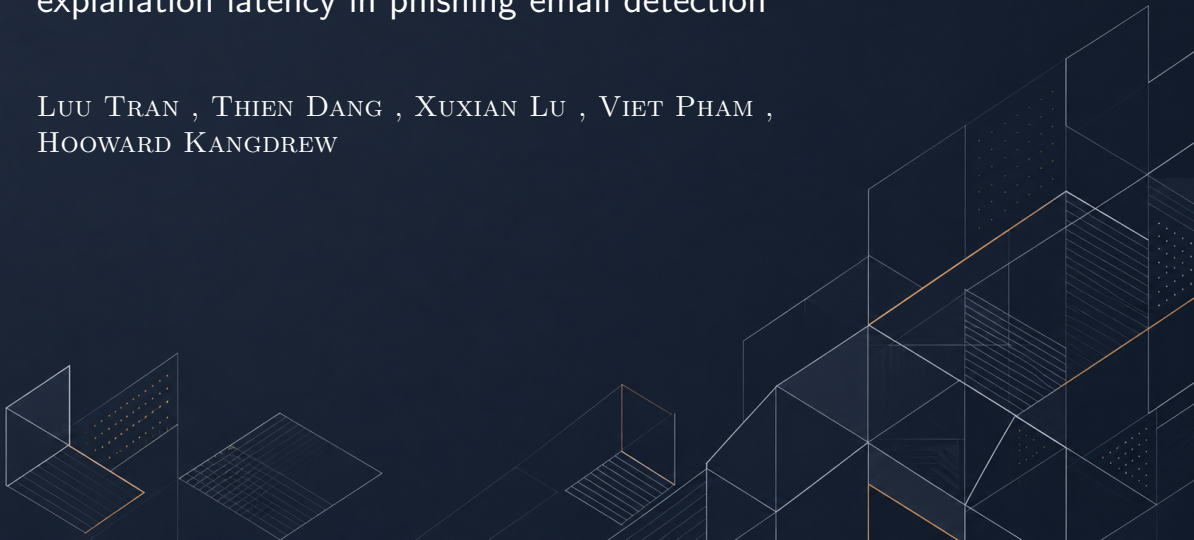


Enhancing global explainability and reducing explanation latency in phishing email detection

LUU TRAN , THIEN DANG , XUXIAN LU , VIET PHAM ,
HOOWARD KANGDREW



1. Problem: Enhancing global explainability and reducing explanation latency in phishing email detection
2. To focus especially on the phishing email detection, the objective is to improve the explainability compared to the current phishing detection models
3. To provides a better and faster solution for business with minimum cost
4. To improve the phishing detection model's credibility and accountability for security and society issues, that phishing detection models must be transparent

1. To help businesses and organizations with a high performance, low cost, and transparent solution with limited resources to understand why the email has been labeled as phishing
2. To help enhance understanding of the behavior for the black box phishing detection model

Authors	Title	XAI methods	AI models
Al-Subaiey et al. (2024)	Novel interpretable and robust web-based AI platform for phishing email detection	LIME	SVM, Multinomial Naive Bayes, Random Forest
Al-E'mari et al. (2025)	A Cognitive BDI-BiLSTM Hybrid Framework for Interpretable Email Threat Detection	SHAP, BDI cognitive agent	BiLSTM
Shendkar et al. (2025)	Enhancing Phishing Attack Detection Using Explainable AI: Trends and Innovations	SHAP, Grad-CAM	Deep learning models

Table: Summary of XAI methods and AI models in phishing email detection studies.

- ▶ Sharma et al. (2025) also identified that LIME and SHAP are 2 most common used method to explain phishing detection model
- ▶ According Hassija et al. (2023), SHAP and LIME are local model agnostic method (local interpreter) which does not explain the model globally

Author	XAI method	Feasible	Risk
Apley & Zhu (2020)	Accumulated Local Effects	Computationally efficient and truly mathematical explanation	Complex interpretations
Friedman & Popescu (2008)	Rule Ensembles	Human-readable logic	Linear assumption and sensitive to threshold
Bastani et al. (2018)	Decision tree surrogate model	Interpretable global surrogate for explaining the original model	Low-depth trees may not fit the training data well

Table: Comparison of selected global explanation methods.

- Instead of decision tree, can we increase model complexity while still remains explainable?
- Therefore we choose family of Generalized Additive Model (GAM) which are intrinsically explainable (Wood, 2017)

- ▶ If we can implement successfully, email phishing detection model can be explained more faithful hence more reliable
- ▶ Faster explanation generation time
- ▶ Help surrogate model technique more robust and position glass-box models.

Practical problem

- ▶ Phishing detection models can be accurate, but **RF and DNN** models are still **difficult for analysts to understand**.
- ▶ Local explainers such as **LIME/SHAP** explain one email at a time, which limits **global understanding** and increases **repeated explanation cost**.

Our focused research problem

Can a faithful global surrogate model explain black-box phishing detectors while reducing the cost of repeated explanation generation?

Integrated public sources

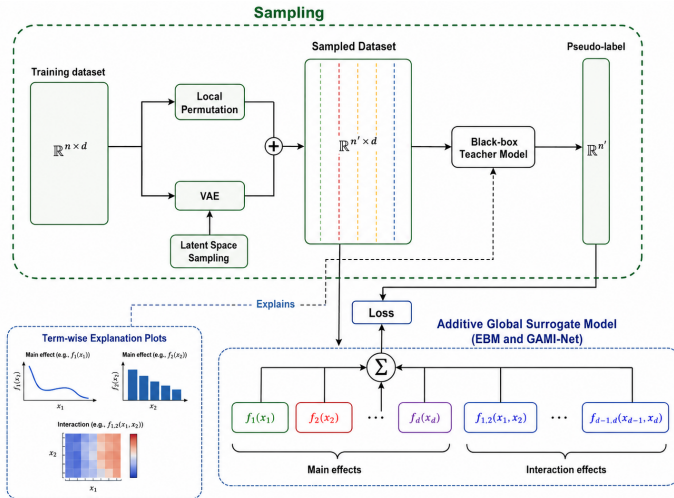
- ▶ **Nazario phishing corpus:** real-world phishing mailbox samples.
- ▶ **Enron Email Dataset:** legitimate workplace emails from about 150 users, about 0.5M messages.
- ▶ **Kaggle phishing email collections:** including Enron / Ling/ CEAS/ Nazario/ Nigerian/ SpamAssassin-based data.

Final dataset outcome

- ▶ **132,289** collected email samples.
- ▶ Label balance: **51.9% phishing** and **48.1% legitimate**. - No bias in model
- ▶ Unified into **17 raw schema fields** before modelling.

Why this supports the research aim

Combining phishing and legitimate business-style emails gives the teacher models enough variation to learn realistic decision patterns, while keeping the dataset publicly sourced and reproducible.



After construct black-box model $F(x)$, we then construct additive global surrogate model $S(x)$

The definition for the Global additive surrogate model is:

$$\text{logit}(S(x)) = \beta_0 + \sum_k f_k(x_k) + \sum_{i < j} f_{ij}(x_i, x_j)$$

Where $f_k(x_k)$ is a term that learn on 1 feature only and $f_{i,j}(x_i, x_j)$ is a term learn on the interaction terms of 2 features.

For EBM, each term is a tree while for GAMI-NET is a sub-neural network

Local perturbation

Apply gaussian noise scaled by standard-deviation of cloud of neighbors to get synthetic samples

$$\tilde{x} = x + \lambda \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_k^2)$$

Variation Auto Encoder Sampling

Use a VAE to learn data distribution and sample from latent space to get synthetic data

$$z \sim \mathcal{N}(0, I), \quad \tilde{x} \sim p_{\theta}(x | z)$$

Black-box model used for experiment: Random Forest and Deep Neural Network

Evaluation metrics

Accuracy Fidelity measures how frequent does the surrogate model agrees with original model

$$\text{Fidelity}_{Acc}(S, F) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\hat{y}_i^S = \hat{y}_i^F \right].$$

F1 Fidelity is the F1-score between the black-box model's predictions and the surrogate model's predictions.

$$\text{Fidelity}_{F1}(S, F) = F_1 \left(\{\hat{y}_i^F\}_{i=1}^N, \{\hat{y}_i^S\}_{i=1}^N \right).$$

Hardware and environment: Intel Core i5 - 14600KF and RTX 4070 super GPU and Python 3.12

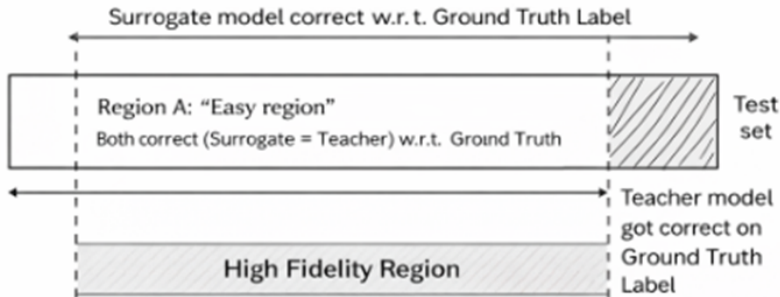


Figure: Surrogate model achieves high fidelity but does not agree with the teacher model on teacher unexpected behavior.

Error Fidelity

Let the error set be defined as:

$$\mathcal{E}_F = \{x_i \mid \hat{y}_i^F \neq y_i\}.$$

Then the error fidelity is defined as:

$$\text{ErrFid}(S, F) = \frac{1}{|\mathcal{E}_F|} \sum_{x_i \in \mathcal{E}_F} \mathbf{1} \left[\hat{y}_i^S = \hat{y}_i^F \right].$$

Teacher	Surrogate	Accuracy	F1
Random Forest	EBM	0.9260	0.9280
Random Forest	GAMI-Net	0.8934	0.8936
Random Forest	Decision Tree Depth 5(Base Line)	0.8311	0.8287
Random Forest	Logistic (Base Line)	<u>0.8013</u>	<u>0.8093</u>
Deep Neural Net	EBM	0.9402	0.9425
Deep Neural Net	GAMI-Net	0.9137	0.9142
Deep Neural Net	Decision Tree Depth 5(Base Line)	0.8491	0.8553
Deep Neural Net	Logistic (Base Line)	<u>0.8093</u>	<u>0.8093</u>

Table: Accuracy and F1 fidelity of each surrogate model with respect to its teacher model.

Teacher	Surrogate	Error Rate	Error Fidelity
Random Forest	EBM	0.0567	0.7673
Random Forest	GAMI-Net	0.0567	0.7367
Random Forest	Decision Tree Depth 5 (Baseline)	0.0567	0.7613
Random Forest	Logistic (Baseline)	0.0567	<u>0.6373</u>
Deep Neural Net	EBM	0.0851	0.7695
Deep Neural Net	GAMI-Net	0.0851	0.7575
Deep Neural Net	Decision Tree Depth 5 (Baseline)	0.0851	<u>0.6510</u>
Deep Neural Net	Logistic (Baseline)	0.0851	0.6803

Table: Error Fidelity comparison across surrogate models.

Teacher	Dataset	EBM ErrFid	GAMI-Net ErrFid
Random Forest	92k raw	0.7227	<u>0.7047</u>
Random Forest	300k perturbation	<u>0.7180</u>	0.7073
Random Forest	300k+VAE	0.7673	0.7367
Deep Neural Net	92k raw	<u>0.5115</u>	<u>0.6811</u>
Deep Neural Net	300k perturbation	0.7194	0.7682
Deep Neural Net	300k+VAE	0.7695	0.7575

Table: Ablation study on different data generation strategies.

Key Insight

- ▶ High fidelity does not guarantee similar model behaviour
- ▶ Two models can agree on most predictions but differ in errors

Evidence from Experiment

- ▶ Random Forest and DNN show high peer fidelity (~ 0.94)
- ▶ But low peer error fidelity
- ▶ Indicates different failure behaviours

Why Error Fidelity Matters

- ▶ Standard fidelity focuses on overall agreement
- ▶ Does not evaluate explanation of incorrect predictions

Our Contribution

- ▶ Proposed global XAI approach for email phishing detection
- ▶ Proposed Error Fidelity to enforce the credibility of the model

Findings

- ▶ Accuracy fidelity > 0.90
- ▶ Error fidelity ≈ 0.75
- ▶ Significant gap between general behaviour and failure cases

Conclusion: Fidelity alone is insufficient to evaluate explanation quality

Model Capability

- ▶ EBM and GAMI-Net achieve high fidelity with teacher models
- ▶ Both outperform baseline models such as Logistic Regression
- ▶ EBM consistently outperforms GAMI-Net

Limitations and future work

- ▶ Gap between fidelity and error fidelity remains
- ▶ Surrogate models do not fully capture unexpected behaviour
- ▶ No unified metric to fully reflect explanation reliability

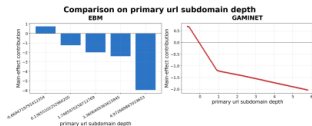


Fig. 5: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on primary subdomain depth

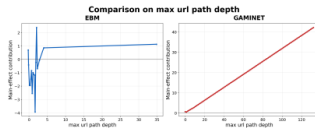


Fig. 6: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on max url path depth

- Shows how features influence phishing prediction
- Positive values increase phishing likelihood
- Negative values reduce phishing likelihood

EBM (Explainable Boosting Machines) and GAMI-Net has been provided that it can effectively approximate complex teaching model and achieving high accuracy rate.

Our approach significantly reduces explanation latency, making real time detection possible for businesses, providing them with a cost-effective and interpretable model to deploy for security protection purposes

Supplemental metrics error fidelity enforce the the credibility of the model

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